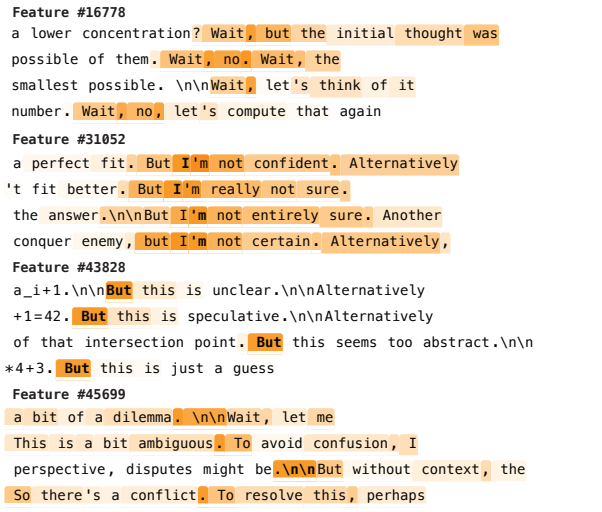
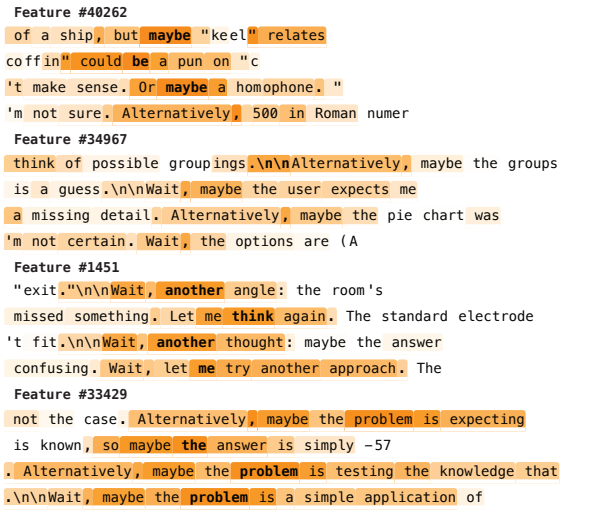
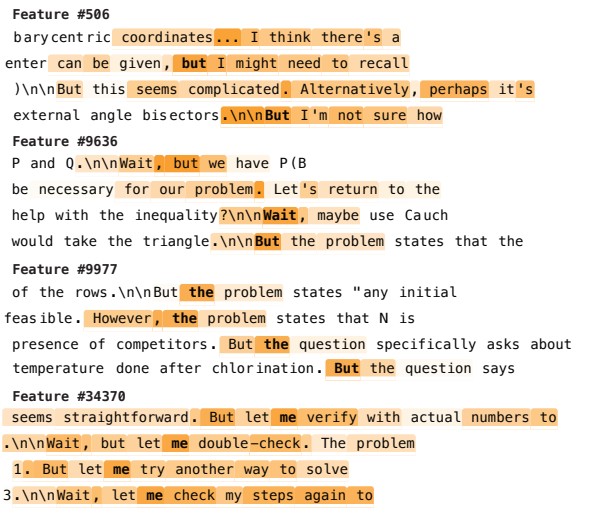



(a) Top-activating examples from the manually verified set of features. The chosen examples represent "uncertainty".

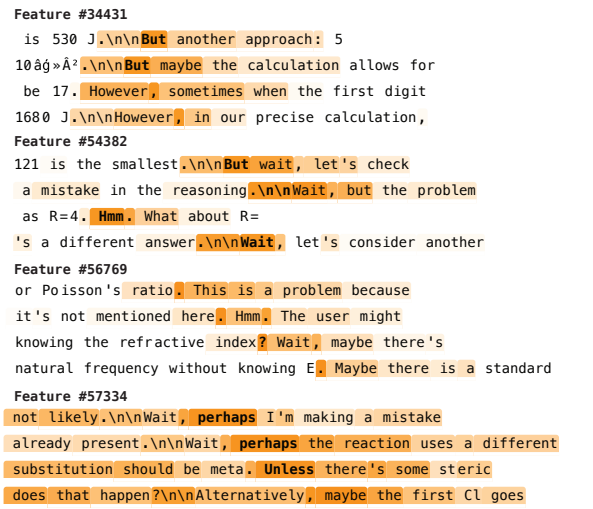



(b) Top-activating examples from the manually verified set of features. The chosen examples represent "exploration".

Figure 8: Examples of feature interfaces used in manual interpretation experiments.




(a) Top-activating examples from the manually verified set of features. The chosen examples represent "reflection".




(b) Top-activating examples from the manually verified set of features. The chosen examples represent mixed behaviors.

Figure 9: Examples of feature interfaces used in manual interpretation experiments.

| Group Name | Features | Possible Function | Effect Type | Observed Behavior |
|---|---|---|---|---|
| Reasoning Depth and Thoroughness | 506, 4395, 9636, 23052, 30288, 33148, 54382, 61935 | Controls multi-step analysis, iteration, and self-correction in problem-solving. | Stylistic & Structural, Semantic & Logical Consistency | **Strengthening**: Extensive step-by-step reasoning, multiple iterations, self-corrections. **Weakening**: Direct answers with minimal steps. |
| Numerical Accuracy and Validation | 4990, 3466, 16778, 46379, 34813, 51765 | Governs precision in calculations, unit conversions, and error-checking. | Semantic & Logical Consistency | **Strengthening**: Meticulous unit tracking, iterative re-evaluation. **Weakening**: Direct results with potential errors. |
| Exploration of Multiple Methods | 22708, 62446 | Encourages evaluating alternative approaches before finalizing solutions. | Semantic & Logical Exploration | **Strengthening**: Compares multiple strategies (e.g., DP vs. greedy). **Weakening**: Commits to the first viable method. |
| Structural and Logical Organization | 57334, 43828, 45699, 49326, 17726, 46449, 41636, 40262 | Ensures clarity, step-by-step breakdown, and logical flow. It may also balance executable code generation vs. verbal explanations. | Structural, Semantic & Instruction Clarity | **Strengthening**: Well-structured explanations. **Weakening**: Disorganized or fragmented reasoning. |
| Symbolic vs. Numerical Reasoning | 48026, 34967, 34589 | Balances theoretical/symbolic reasoning with direct numerical computation. | Semantic & Logical Consistency | **Strengthening**: Algebraic/theoretical frameworks. **Weakening**: Immediate numerical substitution. |
| Self-Correction and Backtracking | 16778, 35337, 42609, 34431, 25953 | Controls iterative refinement and error-checking. | Semantic & Logical Consistency | **Strengthening**: Multiple rounds of self-correction. **Weakening**: Commits to initial answers without revision. |
| Causal Chaining & Scientific Context | 56769, 34370, 3261, 13457 | Enforces clear multi-step causal linkages in science/environmental scenarios, modulates temporal reasoning, hypothetical alternatives and scenario simulation | Semantic (Causality) | **Strengthening**: yields explicit causal chains, regulates contrastive reasoning, gives clearer timeline-based reasoning. **Weakening** results in loosely linked assertions or missing intermediate steps, omit historical or causal context |
| Edge Case and Constraint Handling | 16343, 46691, 3942 | Ensures validation of edge cases and constraints. | Semantic & Logical Consistency | **Strengthening**: Explicitly addresses edge cases. **Weakening**: Assumes valid inputs without verification. |
| Semantic Elaboration & Conceptual Depth | 1451, 33429, 61104, 25485, 45981, 31052, 16441, 53560 | Shapes depth of domain-specific explanations, analogies, trade-off discussions, and interdisciplinary links | Semantic (Elaboration & Breadth) | **Strengthening**: Adds conceptual depth through analogies, trade-offs, and multi-factor explanations. **Weakening**: Reduces to simple, surface-level or single-cause statements with minimal abstraction. |
| Other Behavior | 48792, 9977, 20046, 12036, 32134 | Include: influences engagement and conversational tone, assertiveness/redundancy/structure in text and terminology | Stylistic & Structural | **Strengthening:** Creates a more formal, robotic style with rigid structure and a high degree of confidence in statements. **Weakening:** Makes the style livelier and more conversational, with informal delivery, varied structure, and a moderate level of confidence that includes elements of uncertainty and flexibility. |

Table 4: Reasoning clusters obtained using GPT-4o. Each cluster is defined by a particular type of reasoning (depth of analysis, numerical checking, code generation), the specific features involved, their hypothesized function in shaping the models' output style and logic, and the observed behaviors that emerge when using the feature steering.

**Problem:** How many r's are in the word strawberry?

**Generation (no steering):** I need to determine how many times the letter "r" appears in the word "strawberry." First, I'll write down the word and break it into individual letters: S, T, R, A, W, B, E, R, R, Y. Next, I'll identify which of these letters are "r": the third, eighth, and ninth letters. Finally, I'll count these "r"s, which gives me a total of $\boxed{three}$.

**Generation (steering):** Okay, so I need to figure out how many r's are in the word "strawberry." Let me think about this step by step. First, I should probably write down the word and break it down into its individual letters. The word is spelled S-T-R-A-W-B-E-R-R-Y. Wait, let me make sure I have that right. Sometimes I get confused with the spelling. I think the correct spelling is "strawberry." So, let me write it out again: S, T, R, A, W, B, E, R, R, Y. Hmm, now that I write it, I see there are two R's after the E. Let me count the letters again to make sure I don't miss any. S (1), T (2), R (3), A (4), W (5), B (6), E (7), R (8), R (9), Y (10). So the word has 10 letters in total. Now, focusing on the R's. The third letter is R, that's one. Then after E, which is the seventh letter, there are two more R's: the eighth and ninth letters. So that's two more. Let me double-check the spelling again to make sure I'm not miscounting. Sometimes I might confuse the letters. I think it's S-T-R-A-W-B-E-R-R-Y, so yes, the two R's are towards the end. Wait, is there another R in the word? Let me think. The word is "strawberry." I think it's spelled with an R after the E, and another R after that. So that's two R's. I don't recall any other R's in the word. Let me go through each letter again: S, T, R, A, W, B, E, R, R, Y. So, yes, the R's are the third, eighth, and ninth letters. Wait, that's three R's. Did I miscount earlier? Wait, no, I think I made a mistake. Let me recount the letters with their positions. Let's write it out:
1. S
2. T
3. R
4. A
5. W
6. B
7. E
8. R
9. R
10. Y

So, the R's are at positions 3, 8, and 9. That makes three R's in total. I think I initially thought it was two, but upon a closer examination, it's actually three. Let me check another source or think about how the word is pronounced. The word "strawberry" is pronounced with a double R sound, so that would correspond to two R's, but when I break it down, I see that the letters are R, R. Wait, no, the letters are E, R, R. So, that's two R's after the E. So, the third letter is R, and then the eighth and ninth letters are R's as well. So that's three R's. Wait, I'm getting confused. Let me visualize the word again: S-T-R-A-W-B-E-R-R-Y. So, the letters are S, T, R, A, W, B, E, R, R, Y. So, the R appears at the third position, and then again at the eighth and ninth positions. That makes three R's. So, the correct count is three R's. I think I initially miscounted because I focused on the double R at the end, but there's another R earlier in the word. So, to clarify, the word "strawberry" has three R's: one in the third position, and two more at the end. So, the correct answer is $\boxed{three}$ R's.

Table 5: An example of "How many r's are in the word strawberry?" problem and corresponding full outputs from DEEPSEEK-R1-LLAMA-8B and its steered version.